文档技术报告

高三（10）杨逍

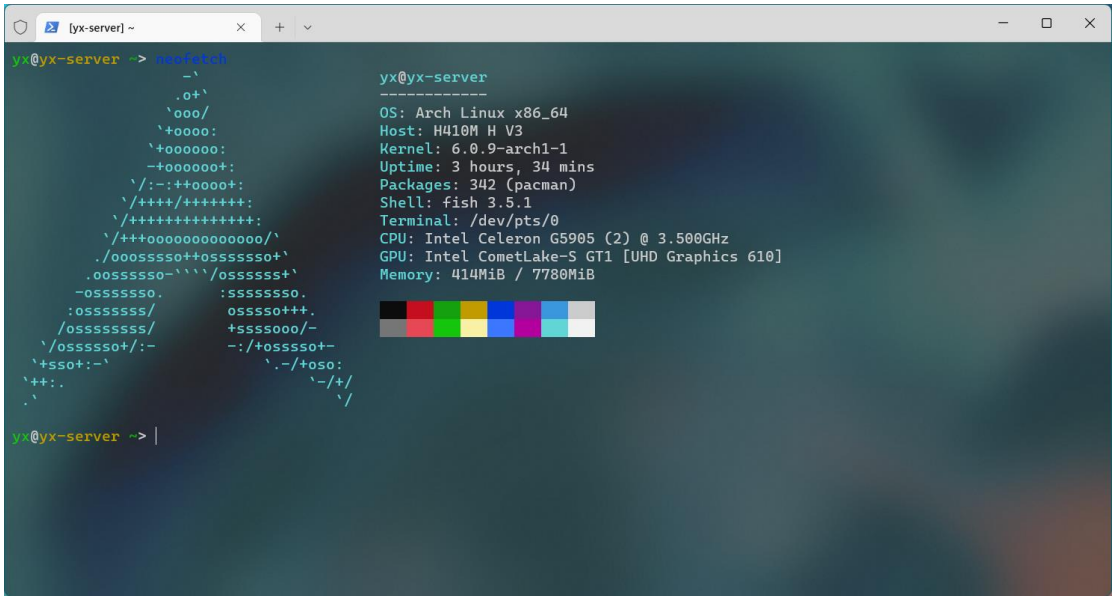
需求：我日常工作环境为 Windows11+Arch Linux，然而在 Arch Linux 上并未找到稳定好用的 QQ 客户端；家中旧打印机 Brother HL-1208 不支持网络打印，平时打印作业必须开电脑，且家庭成员不能共用打印机，带来麻烦；大量摄影作品原片塞满硬盘，希望可以通过网络存储实现便利访问同时解放本机存储空间。

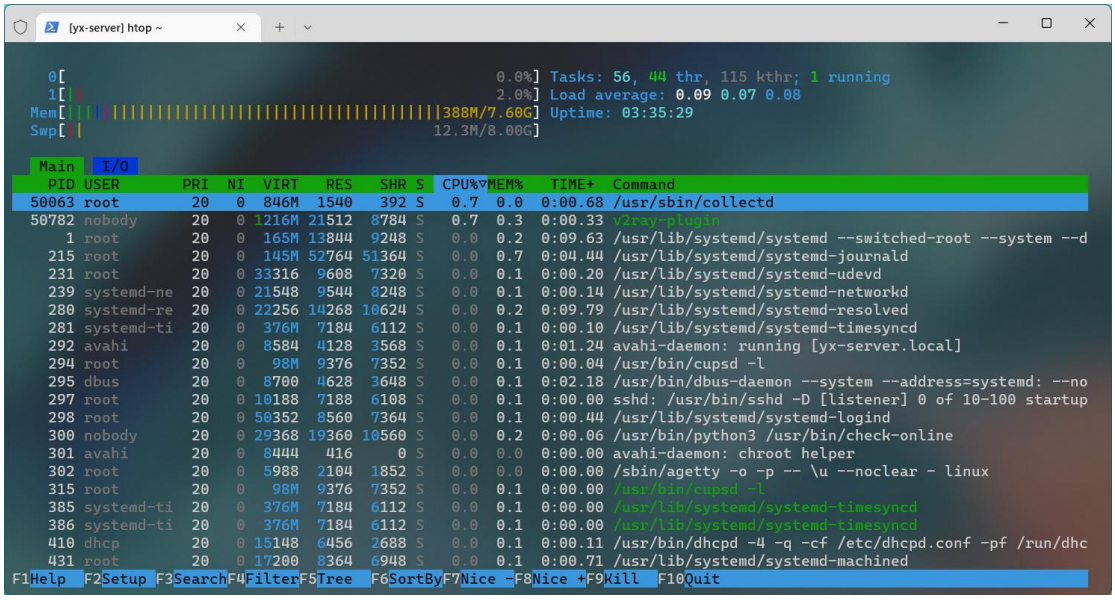
硬件：这些需求对硬件要求都不高，因此选用廉价的 H410M 主板+G5905 处理器搭建服务器，NAS 硬盘选用 WD Red Plus 2TB，整机总价约￥1300

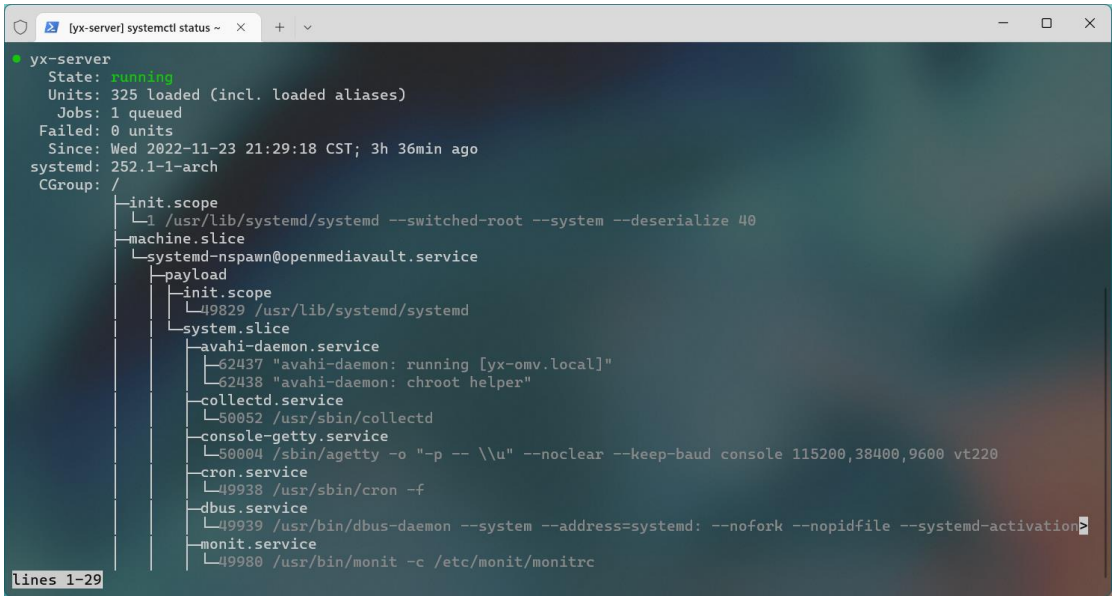
软件：由于我日常使用最熟悉的 Linux 系统即 Arch Linux，而且 Arch Wiki 对于各项服务的配置可以起到很好的参考与帮助作用，以及 Arch Linux 强大的通用性，因此选择 Arch Linux 作为家庭服务器系统

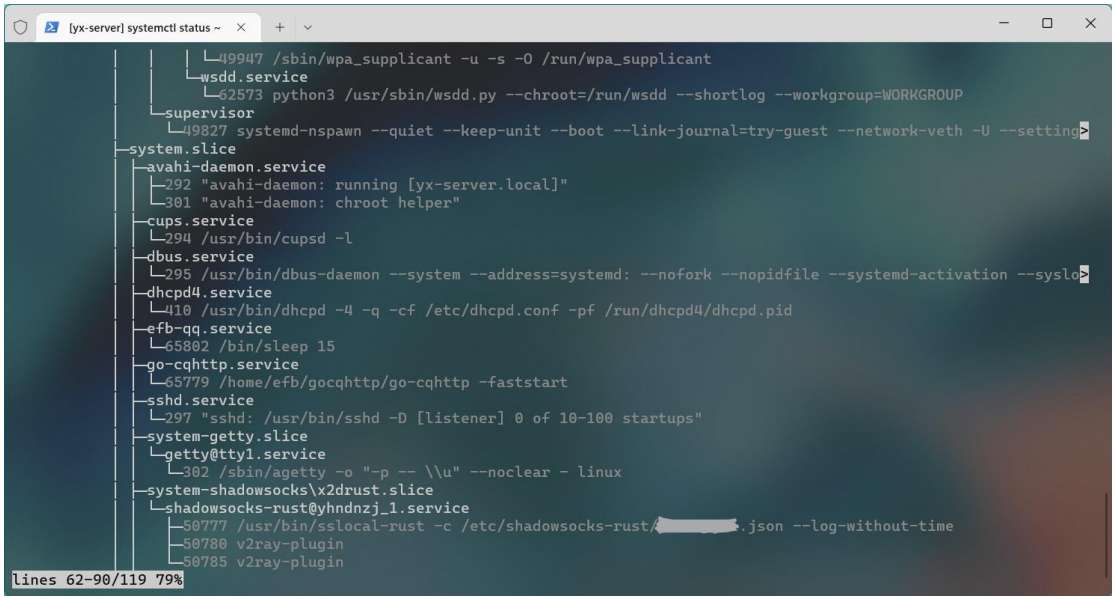
各项具体服务配置：由于此文档在 Windows 上撰写，故使用 Windows 终端 SSH 连接服务器进行截图展示

1. 系统概况




[yx-server] htop ~
0[0.0%] Tasks: 56, 44 thr, 115 kthr; 1 running
1[2.0%] Load average: 0.09 0.07 0.08
Mem[388M/7.60G] Uptime: 03:35:29
Swp[12.3M/8.00G]
Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
50063 root 20 0 846M 1540 392 S 0.7 0.0 0:00.68 /usr/sbin/collectd
50782 nobody 20 0 1216M 21512 8784 S 0.7 0.3 0:00.33 v2ray-plugin
1 root 20 0 165M 13844 9248 S 0.0 0.2 0:09.63 /usr/lib/systemd/systemd --switched-root --system --d
215 root 20 0 145M 52764 51364 S 0.0 0.7 0:04.44 /usr/lib/systemd/systemd-journald
231 root 20 0 33316 9608 7320 S 0.0 0.1 0:00.20 /usr/lib/systemd/systemd-udevd
239 systemd-ne 20 0 21548 9544 8248 S 0.0 0.1 0:00.14 /usr/lib/systemd/systemd-networkd
280 systemd-re 20 0 22256 14268 10624 S 0.0 0.2 0:09.79 /usr/lib/systemd/systemd-resolved
281 systemd-ti 20 0 376M 7184 6112 S 0.0 0.1 0:00.10 /usr/lib/systemd/systemd-timesyncd
292 avahi 20 0 8584 4128 3568 S 0.0 0.1 0:01.24 avahi-daemon: running [yx-server.local]
294 root 20 0 98M 9376 7352 S 0.0 0.1 0:00.04 /usr/bin/cupsd -l
295 dbus 20 0 8700 4628 3648 S 0.0 0.1 0:02.18 /usr/bin/dbus-daemon --system --address=systemd: --no
297 root 20 0 10188 7188 6108 S 0.0 0.1 0:00.00 sshd: /usr/bin/sshd -D [listener] 0 of 10-100 startup
298 root 20 0 50352 8560 7364 S 0.0 0.1 0:00.44 /usr/lib/systemd/systemd-logind
300 nobody 20 0 29368 19360 10560 S 0.0 0.2 0:00.06 /usr/bin/python3 /usr/bin/check-online
301 avahi 20 0 8444 416 0 S 0.0 0.0 0:00.00 avahi-daemon: chroot helper
302 root 20 0 5988 2104 1852 S 0.0 0.0 0:00.00 /sbin/agetty -o -p -- \u --noclear - linux
315 root 20 0 98M 9376 7352 S 0.0 0.1 0:00.00 /usr/bin/cupsd -l
385 systemd-ti 20 0 376M 7184 6112 S 0.0 0.1 0:00.00 /usr/lib/systemd/systemd-timesyncd
386 systemd-ti 20 0 376M 7184 6112 S 0.0 0.1 0:00.00 /usr/lib/systemd/systemd-timesyncd
410 dhcp 20 0 15148 6456 2688 S 0.0 0.1 0:00.11 /usr/bin/dhcpd -4 -q -cf /etc/dhcpd.conf -pf /run/dhc
431 root 20 0 17200 8364 6948 S 0.0 0.1 0:00.71 /usr/lib/systemd/systemd-machined
F1Help F2Setup F3SearchF4FilterF5Tree F6SortByF7Nice -F8Nice +F9Kill F10Quit

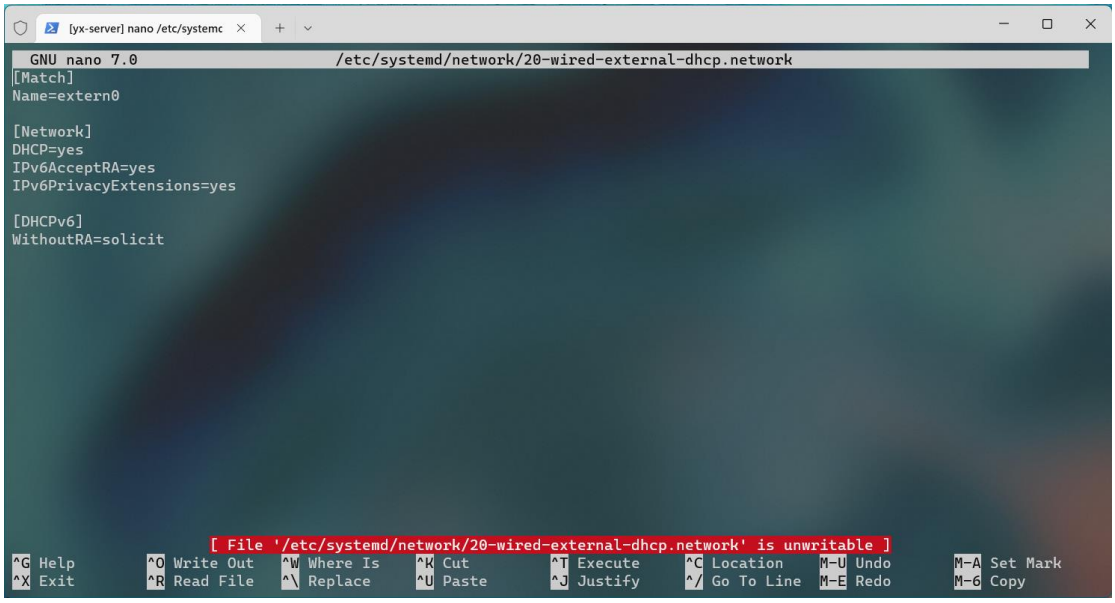

[yx-server] systemctl status ~
yx-server
State: running
Units: 325 loaded (incl. loaded aliases)
Jobs: 1 queued
Failed: 0 units
Since: Wed 2022-11-23 21:29:18 CST; 3h 36min ago
systemd: 252.1-1-arch
CGroup: /
├─init.scope
│ └─1 /usr/lib/systemd/systemd --switched-root --system --deserialize 40
├─machine.slice
│ └─systemd-nspawn@openmediavault.service
│ ├─payload
│ │ ├─init.scope
│ │ │ └─49829 /usr/lib/systemd/systemd
│ │ └─system.slice
│ │ ├─avahi-daemon.service
│ │ │ ├─62437 "avahi-daemon: running [yx-omv.local]"
│ │ │ └─62438 "avahi-daemon: chroot helper"
│ │ ├─collectd.service
│ │ │ └─50052 /usr/sbin/collectd
│ │ ├─console-getty.service
│ │ │ └─50004 /sbin/agetty -o "-p -- \\u" --noclear --keep-baud console 115200,38400,9600 vt220
│ │ ├─cron.service
│ │ │ └─49938 /usr/sbin/cron -f
│ │ ├─dbus.service
│ │ │ └─49939 /usr/bin/dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation>
│ │ ├─monit.service
│ │ │ └─49980 /usr/bin/monit -c /etc/monit/monitrc
lines 1-29


[yx-server] systemctl status ~
│ │ └─49947 /sbin/wpa_supplicant -u -s -O /run/wpa_supplicant
│ └─wsdd.service
│ └─62573 python3 /usr/sbin/wsdd.py --chroot=/run/wsdd --shortlog --workgroup=WORKGROUP
└─supervisor
└─49827 systemd-nspawn --quiet --keep-unit --boot --link-journal=try-guest --network-veth -U --setting>
├─system.slice
│ ├─avahi-daemon.service
│ │ ├─292 "avahi-daemon: running [yx-server.local]"
│ │ └─301 "avahi-daemon: chroot helper"
│ ├─cups.service
│ │ └─294 /usr/bin/cupsd -l
│ ├─dbus.service
│ │ └─295 /usr/bin/dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslo>
│ ├─dhcpd4.service
│ │ └─410 /usr/bin/dhcpd -4 -q -cf /etc/dhcpd.conf -pf /run/dhcpd4/dhcpd.pid
│ ├─efb-qq.service
│ │ └─65802 /bin/sleep 15
│ ├─go-cqhttp.service
│ │ └─65779 /home/efb/gocqhttp/go-cqhttp -faststart
│ ├─sshd.service
│ │ └─297 "sshd: /usr/bin/sshd -D [listener] 0 of 10-100 startups"
│ ├─system-getty.slice
│ │ └─getty@tty1.service
│ │ └─302 /sbin/agetty -o "-p -- \\u" --noclear - linux
│ ├─system-shadowsocks\x2drust.slice
│ │ └─shadowsocks-rust@yhndnzj_1.service
│ │ ├─50777 /usr/bin/sslocal-rust -c /etc/shadowsocks-rust/[redacted].json --log-without-time
│ │ ├─50780 v2ray-plugin
│ │ └─50785 v2ray-plugin
lines 62-90/119 79%

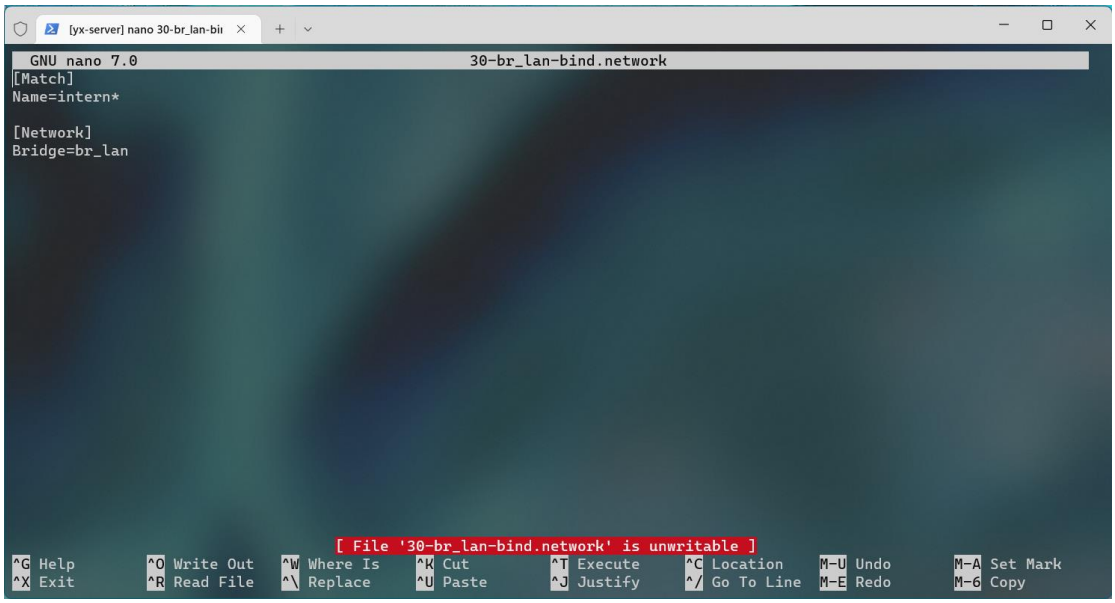
System Status 截取了主要两页（Shadowsocks 使用了同学的代理服务器）
正常工作时 htop 监测 CPU 占用<10%，内存总占用<1GB

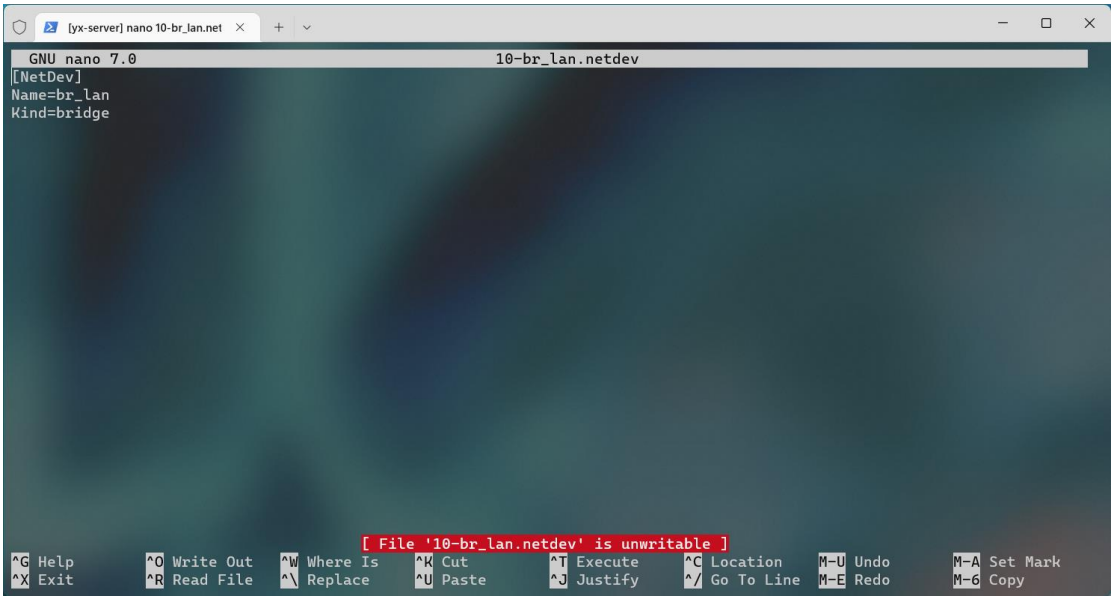
2. 软路由

参考文档 https://yhndnzj.com/2021/08/13/arch-is-the-best-router/
几个配置文件

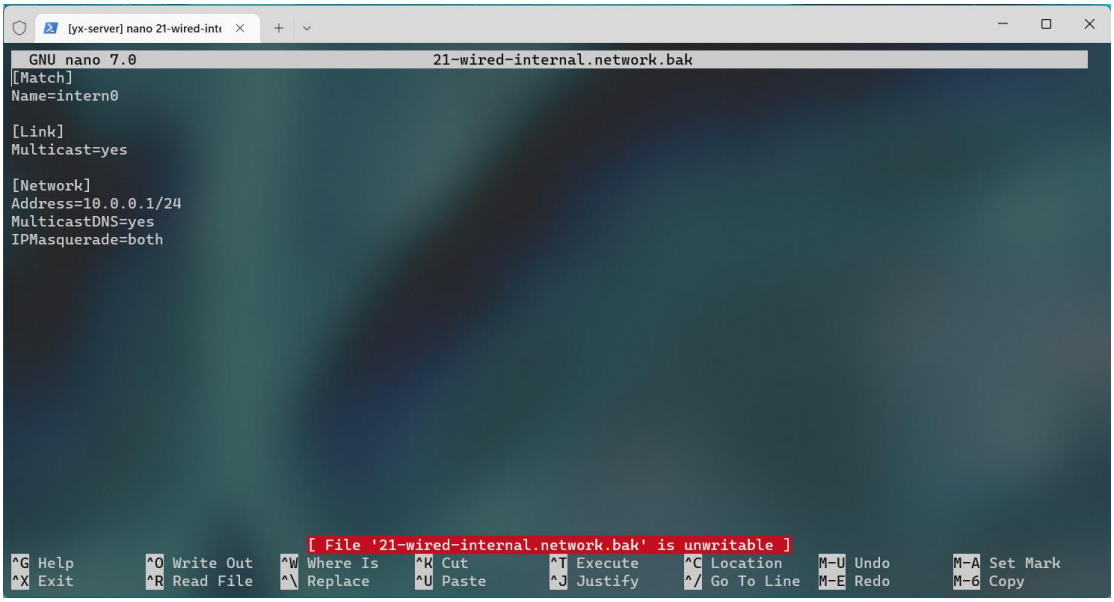


WAN

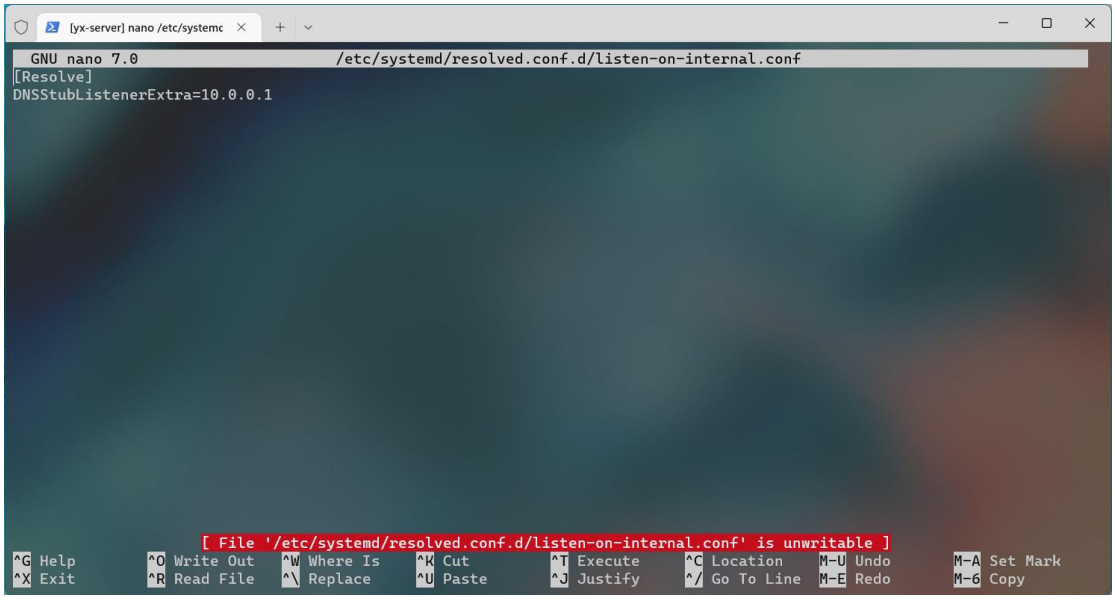



[yx-server] nano 10-br_lan.net
GNU nano 7.0 10-br_lan.netdev
[NetDev]
Name=br_lan
Kind=bridge
[File '10-br_lan.netdev' is unwritable]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy

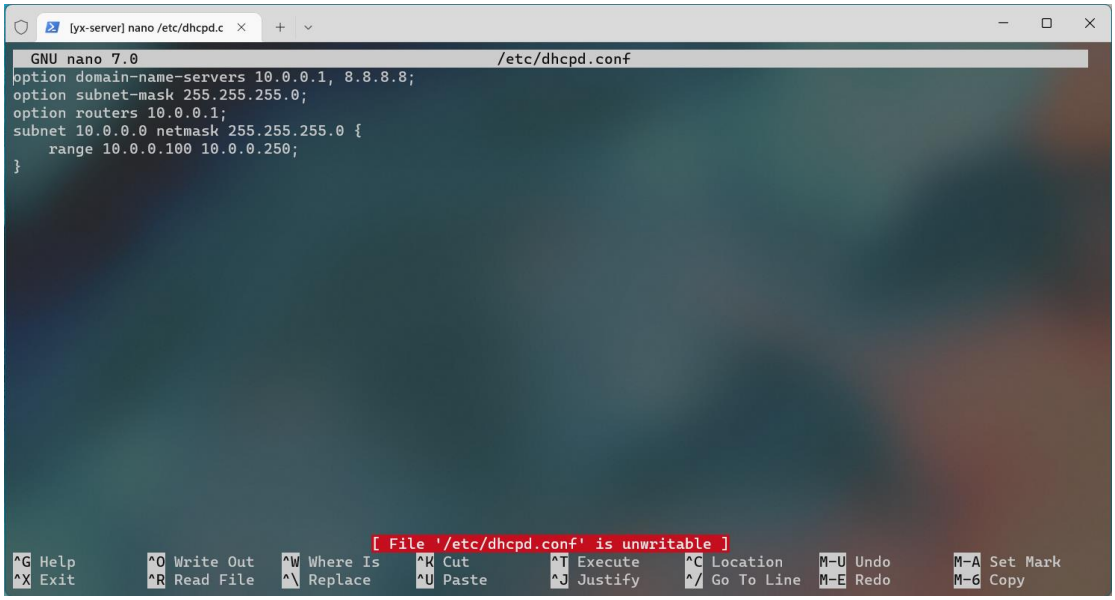
Bridge


[yx-server] nano 21-wired-int
GNU nano 7.0 21-wired-internal.network.bak
[Match]
Name=intern0
[Link]
Multicast=yes
[Network]
Address=10.0.0.1/24
MulticastDNS=yes
IPMasquerade=both
[File '21-wired-internal.network.bak' is unwritable]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy

LAN



DNS（systemd-resolved）



dhcpd

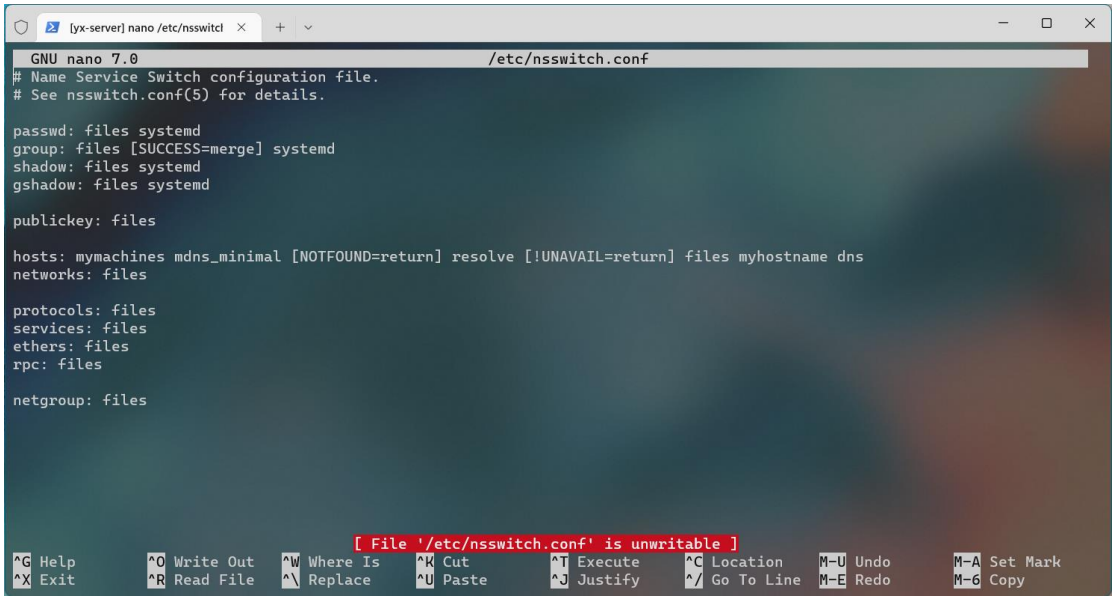
完成配置之后修改 systemd.service 并启动所有服务

以及一些网络优化，参考 https://wiki.archlinux.org/title/Sysctl 中的 Networking 部分

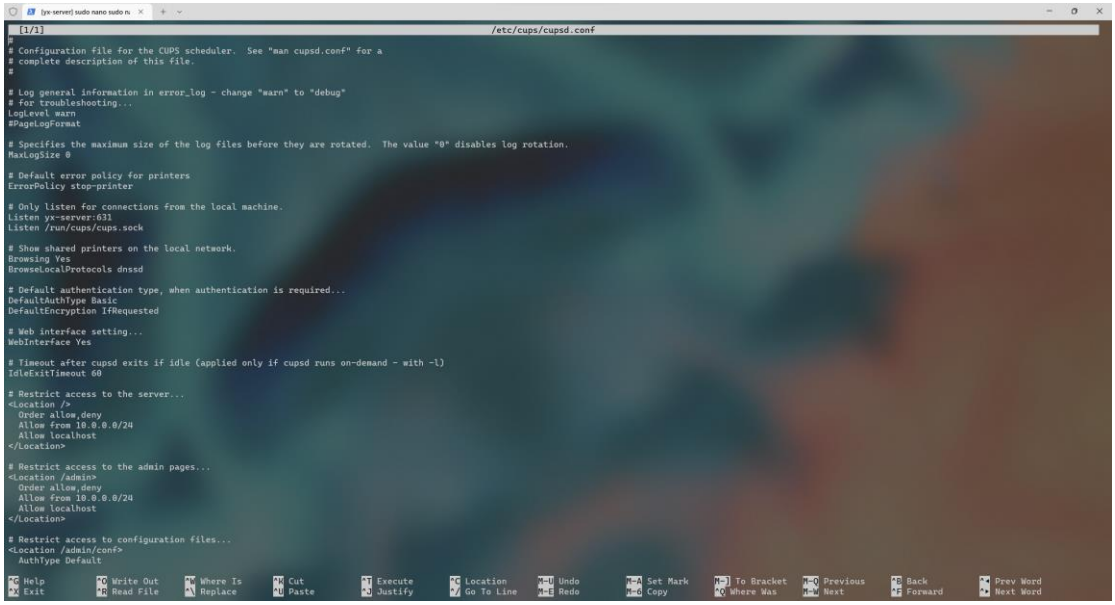
3. CUPS 无线打印



最终通过 CUPS 网络面板实现管理
使用驱动包 brlaser

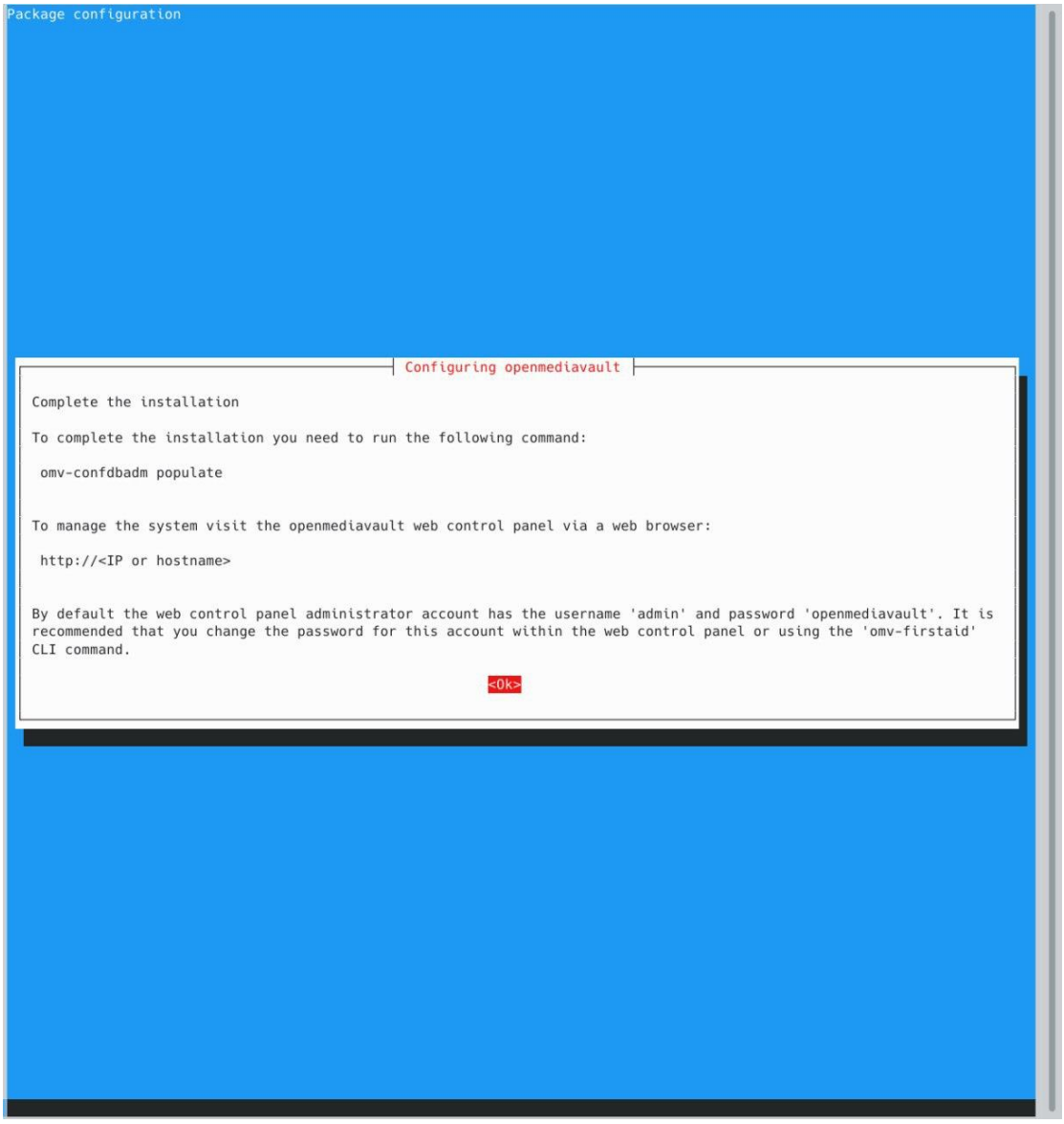


由于 CUPS 强依赖于 Avahi，故使用 Avahi 进行域名解析

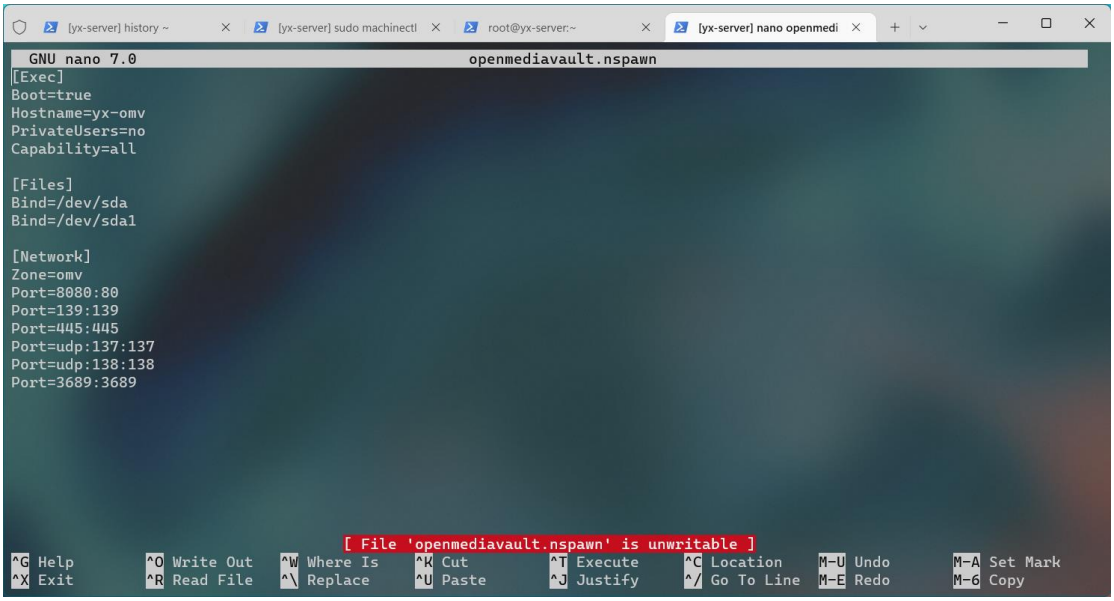


配置 CUPS 以允许本地访问
其余打印机设置在 CUPS 图形化面板中完成
Windows 直接使用，Linux 先删除本地 CUPS 后正常使用

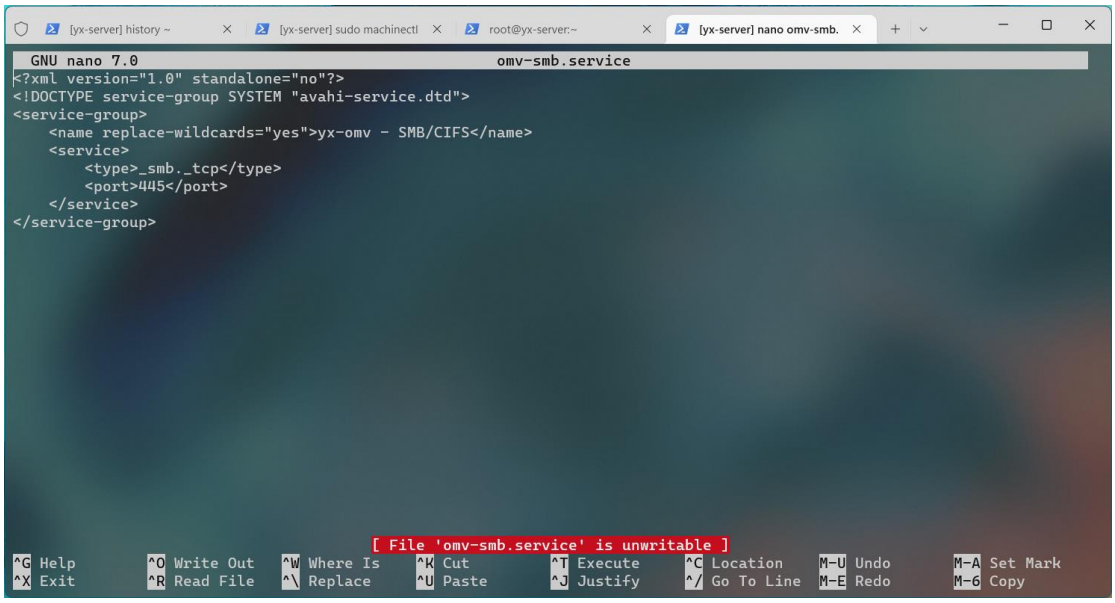
4. OpenMediaVault 网络存储



使用 debootstrap 创建 Debian 环境并拉取镜像进行安装

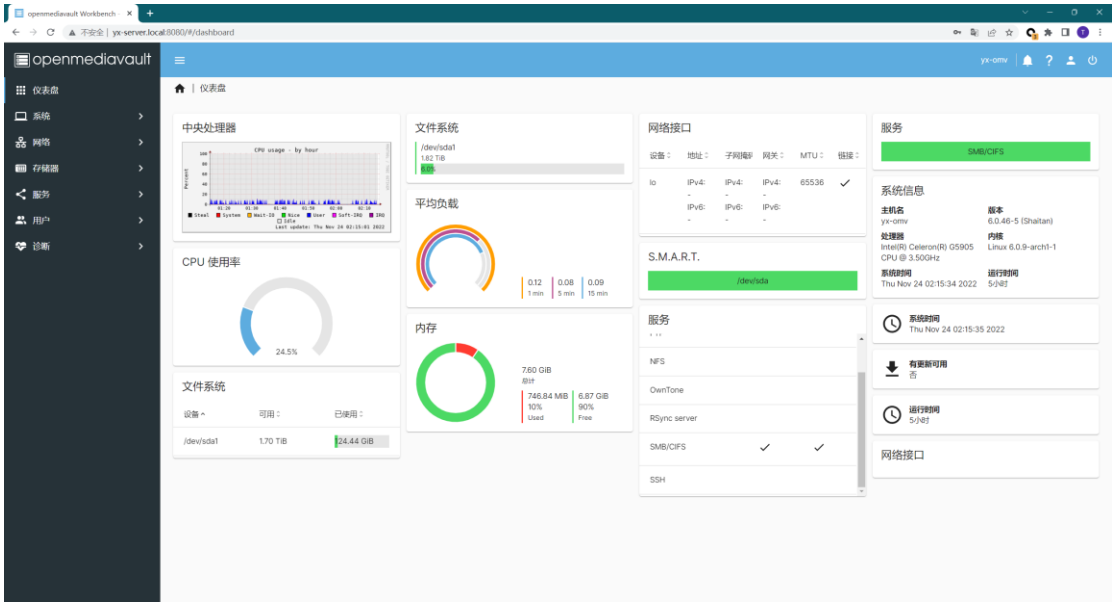


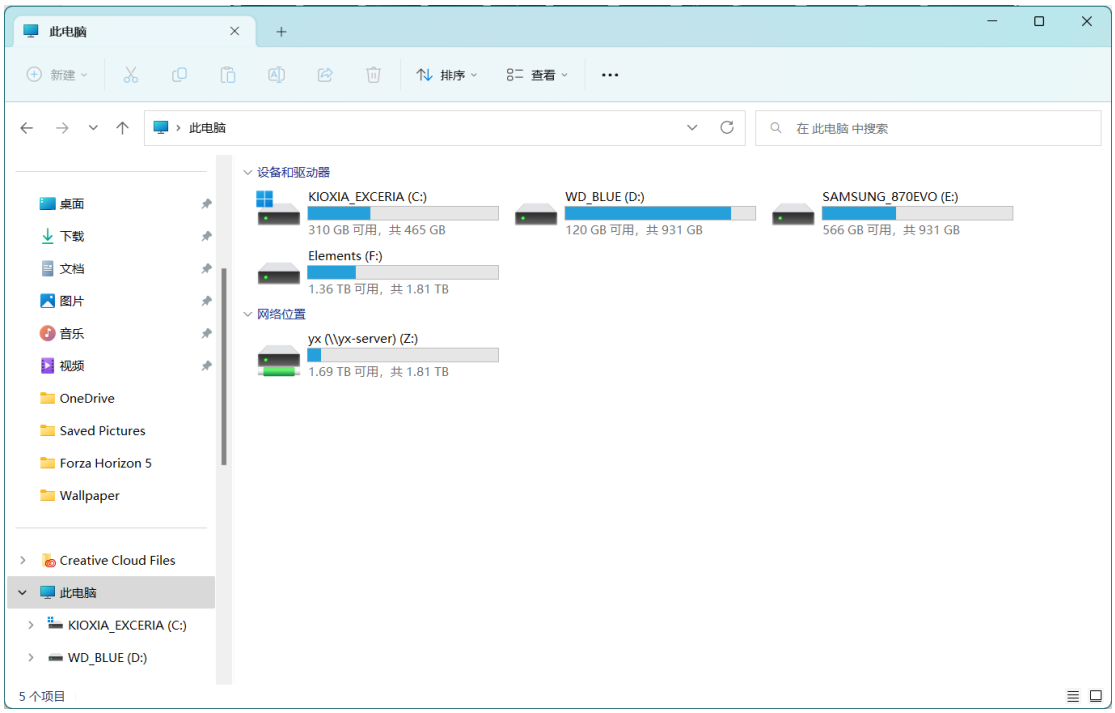
nspawn 容器配置文件



avahi 配置文件

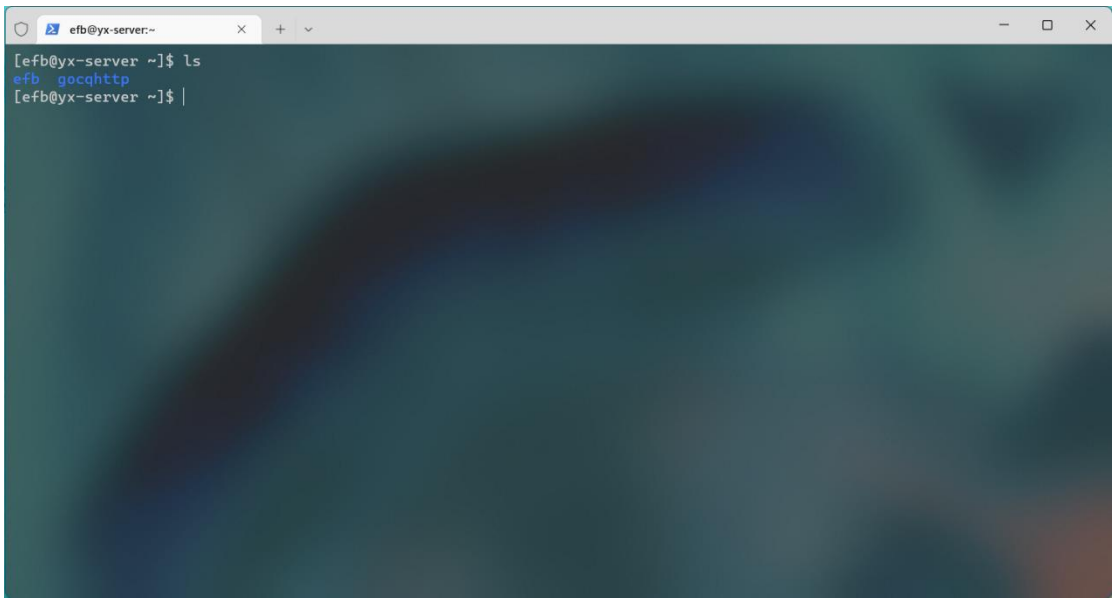
最终使 dolphin 能够自动探测网络存储，~~但 Windows 仍需要手动连接~~

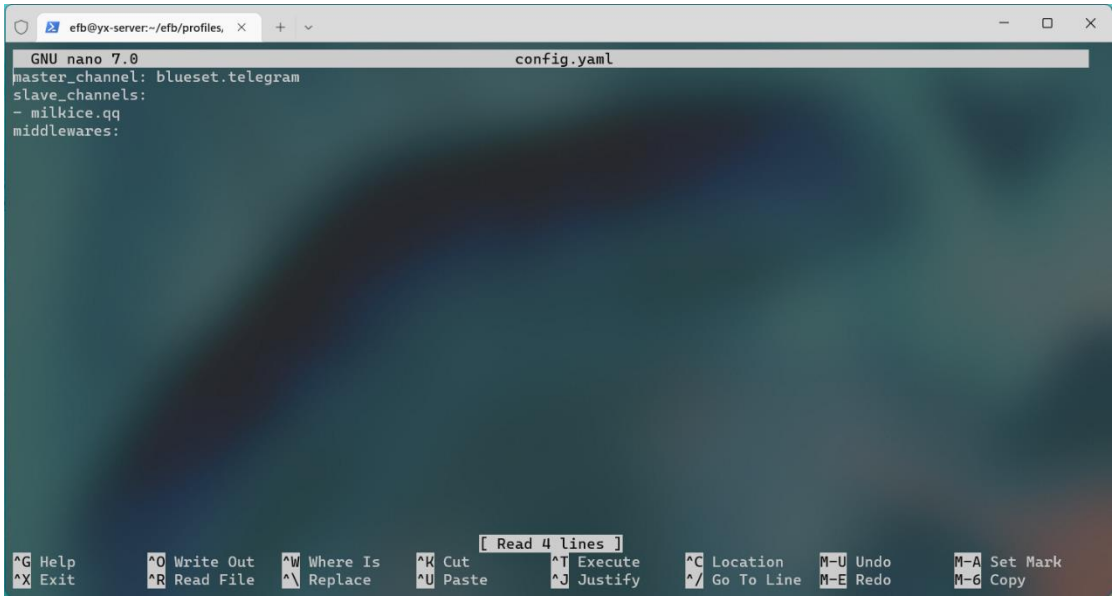




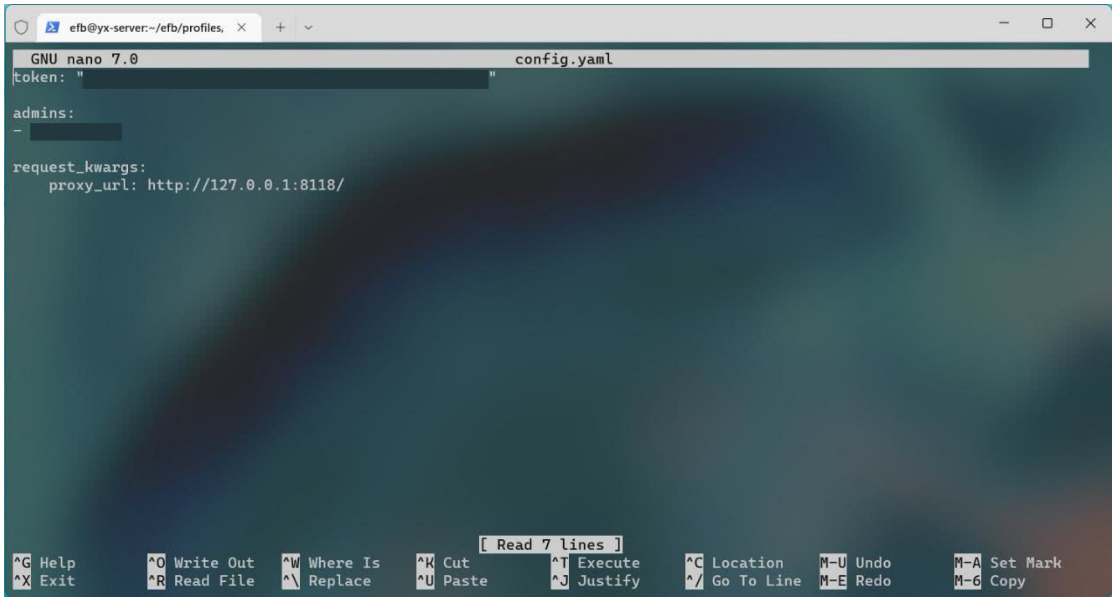
5. EH Forwarder Bot

参考文档：https://ehforwarderbot.readthedocs.io/en/latest/getting-started.html
https://milkice.me/2018/09/17/efb-how-to-send-and-receive-messages-from-qq-on-telegram/
https://github.com/ehForwarderBot/efb-telegram-master
为了防止家目录混乱，新建了一个用户

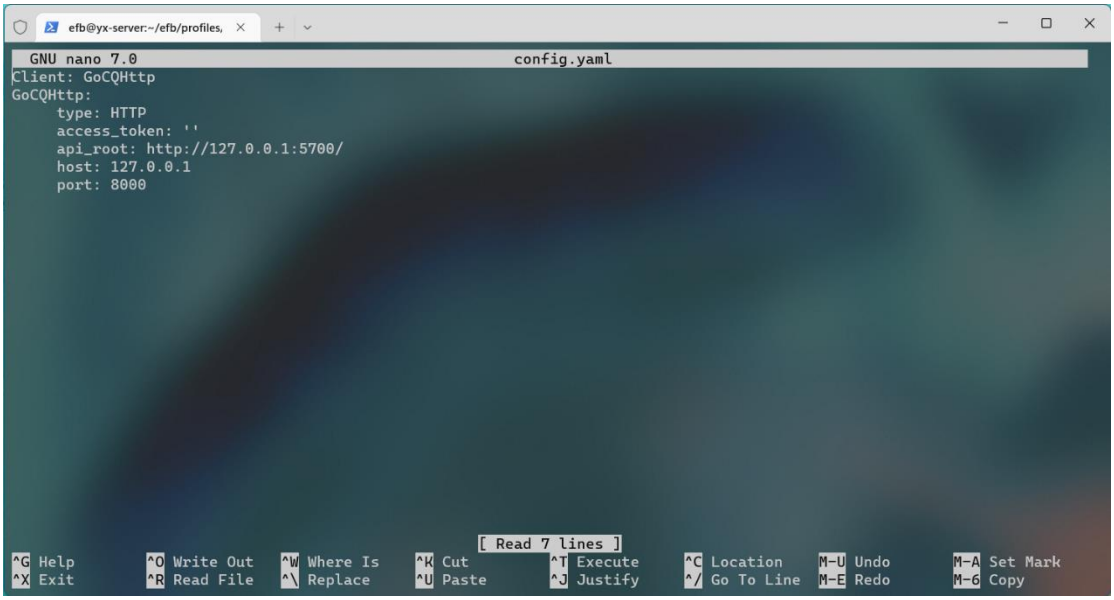




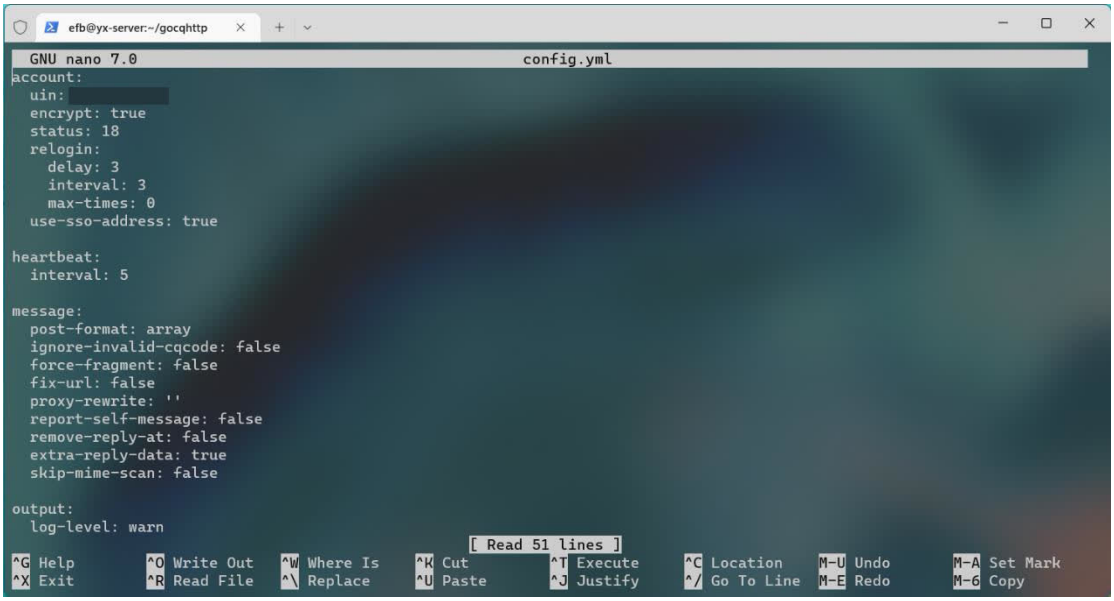
EFB 配置文件



Telegram 主端配置文件



EFB 从端配置



go-cqhttp 配置